

Question # 3 of 5 (Start time: 04:12:30 PM, 13 November 2025)

If $Z = A' + B + C' + D$ then what should be the values of A, B, C, and D so that the expression returns $Z = 0$.

Select the correct option

☐	A = 1, B = 0, C = 0, D = 0
☐	A = 1, B = 0, C = 1, D = 1
☒	A = 1, B = 0, C = 1, D = 0
☐	A = 0, B = 1, C = 0, D = 0

Question # 1 of 5 (Start time: 04:11:55 PM, 13 November 2025)

According to the distributive law of Boolean algebra, which of the following statements is **TRUE**?

Select the correct option

☐	$X + (Y Z) = XY + XZ$
☒	$X + (Y Z) = (X + Y) (X + Z)$
☐	$X + (Y Z) = XYZ$
☐	$X + (Y Z) = X + Y + Z$

Question # 3 of 5 (Start time: 04:58:51 PM, 13 November 2025)

_____ is also known as a Universal gate.

Select the correct option

☐	AND
☒	NOR
☐	OR
☐	XNOR

Question # 4 of 5 (Start time: 04:12:51 PM, 13 November 2025)

The Boolean function $Z = A + B \cdot C'$ can be implemented in digital logic circuit by connecting _____ gates together.

Select the correct option

☒	3
☐	1
☐	4
☐	2

Question # 2 of 5 (Start time: 04:12:09 PM, 13 November 2025)

DE Morgan's Theory is used to convert _____ gates to _____ ones.

Select the correct option

☐	AND/XOR, XOR/AND
☐	OR/NOT, NOT/OR
☒	AND/OR, NOR/NAND
☐	AND/NAND, OR/NOR